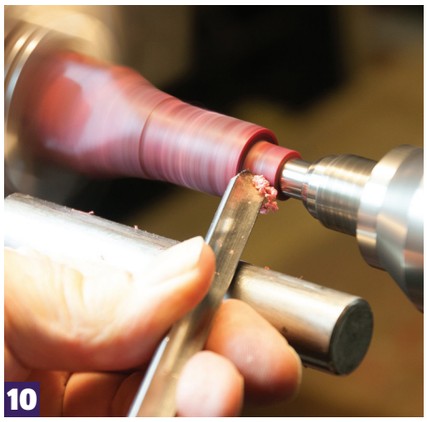
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end of the barrel to accept the cap. Don't worry too much at this stage about shaping just turn down until the end where the spigot will be is cylindrical

10 Now turn the spigot to the exact size required for the die that you will use to cut the thread. Once again check with the specification of the die to get the correct dimensions. I used a combination of a spindle gouge and beading and parting tool to make the cuts and a digital Vernier caliper to carefully measure the size

11 In order for the cap to screw up fully clearance is required behind the threads. Using a thin parting tool I cut a narrow groove to the same depth as the thread – i.e. the difference between the major and minor diameters of the thread. Care is needed not to cut too deep or you will at best weaken the structure of the component at at worst part it off altogether

12 The thread is cut using a die mounted in a die holder in exactly the same way as for the cutting of the threads for the section in last month's article. A chamfer on the end of the spigot will help to start the die. The die I used was an adjustable split die so I was able to cut the thread in stages to reduce the stress on the material and allow a fine final cut to give a good clean thread, but whatever sort of die you use, care is required so slow gentle progress is the way to go

13 The blank was now removed from the chuck and a mandrel created to fit its internal thread. I made this one from ebonite using the same die used to create the threads on the section

14 The blank can now be reversed and re-mounted on the mandrel and supported by a revolving centre in the tailstock in order for the body to be shaped

15 All of the shaping and finishing of the barrel can be done while mounted on this mandrel except for the end which is finished by hand after parting off. Parting off is best done by cutting almost through with a parting tool and then removing from the lathe and making the final cut with a razor saw or sharp craft knife

16 After a final polish on the buffing wheel the barrel is complete, the section can be fitted and we can move on to making the cap

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